

Darwin Smith

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EDUCATION

University of California, San Diego

B.S. in Mathematics and Computer Science

San Diego, CA

June 2026

EXPERIENCE

Software Engineer & Product Designer

Nov 2025 – Present

Veraflux (veraflux.org), self-funded, pre-launch health-research product

San Diego, CA

- Built Veraflux solo: it writes cited supplement reports from PubMed studies for a given age, sex, and goal, and every change ships through 25 CI check scripts (React/TypeScript, Node/Express, FastAPI, PostgreSQL, Stripe).
- LLM pipeline (RAG):** 13 stages per uncached report across Gemini and Claude. Parallel efficacy and safety searches scan up to 470 PubMed candidates; one report cited 52 of 229 scanned, and outside citations are dropped.
- Label scanning:** a single Gemini vision call (\$0.0017) reads a supplement label, including proprietary-blend amounts. Each ingredient gets a cited brief costing \$0.018, about 1/24th of a full report, cached for every user.
- Cost tracking:** logged the cost of every model call, found a fully uncached report costs \$0.43, and matched a day's Anthropic invoice exactly. Limited free-tier reports to 165 cache keys no matter how many people sign up.
- Wrote quota claims as one atomic SQL statement with refunds, made Stripe's 9 webhook event types idempotent, and added 25 CI check scripts, one catching 8 async routes (checkout included) that hang on errors.
- Built a self-hosted BI suite inside the product's own Postgres: 42 endpoints behind 18 admin pages, a 37-event schema with per-event PII allowlists, and hand-written funnel and retention SQL.
- Added a version check so a server deploy cannot set off regeneration of every cached report before the pipeline can build the new version. The API also sheds load before it charges quota or money.
- LLM deadlines:** a request's time budget starts when it arrives. Each model call and retry gets only the time left, and work that sat in the queue past its budget is refused before it costs anything.
- Closed an XXE/SSRF path in PubMed XML parsing with a host allowlist. When its first version missed an NIH subdomain and broke retrieval, added a 503 error so outages no longer read as "No studies found".
- Found the production database could not be rebuilt from its migrations. Wrote baseline migrations, replayed all 31 on a fresh staging database, and confirmed its columns and indexes matched production.
- Prerendered 45 public routes with headless Chromium and generated a 237-URL sitemap for search.
- Frontend:** report scrolling re-rendered the entire report every frame, about 25,500 regex runs per render; fixed it. Also built the component that turns statistics and terms in AI text into accessible popovers.

Test Engineer

Nov 2023 – Apr 2026

SiteHive

San Diego, CA

- Ran US QA and final validation for IoT deployments as technical contact for major enterprise clients.
- Diagnosed firmware and network faults on Hexanode devices with TeraTerm and STM32CubeProgrammer, and tightened validation around recurring hardware defects.

PROJECTS

Club Triton (Collection Strategy Game) | JavaScript, IndexedDB, GitHub Actions

Apr 2025 – June 2025

- Built the game loop and deck-building system (IndexedDB) and merged team work through code reviews.
- Set up CI/CD in GitHub Actions that ran builds, ESLint, and Playwright regression tests on every push.

Global Conflict & Defense Stock Analysis | Python, Pandas, Seaborn

Sep 2025 – Dec 2025

- Merged 20+ years of stock and conflict data; found $r = 0.67$ between conflict deaths and defense valuations.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Java, C/C++, C#, Bash

AI & Data: LLM APIs (Gemini, Claude), retrieval-augmented generation (RAG), multimodal vision, structured output, prompt engineering, Batch API, Pandas, NumPy, Scikit-Learn

Frontend: React, Vite, HTML/CSS, Framer Motion

Backend & Databases: Node/Express, FastAPI, REST APIs, PostgreSQL (Prisma), Zod, Stripe, Supabase

Cloud, DevOps & Tooling: Docker, Railway, GitHub Actions, AWS, Azure, Git, Playwright, Jest, JUnit